Remote Sensing Teacher: Cross-Domain Detection Transformer With Learnable Frequency-Enhanced Feature Alignment in Remote Sensing Imagery

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***Abstract*— Unsupervised domain adaptation (UDA) is critical for remote sensing object detection in real applications, aiming to address the significant performance degradation issue caused by the domain gap between the source and target domain. This method achieves cross-domain alignment by leveraging the unlabeled target domain data, thus avoiding the expensive annotation cost. However, existing works mainly cope with convolutional neural network (CNN)-based object detectors, which are characterized by complex adversarial learning architecture and fail to accurately align the features in remote sensing images with sparsely allocated objects and inevitable background noise. Compared to CNN-based methods, the detection transformer (DETR) largely simplifies the object detection pipeline and demonstrates the great potential of its intrinsic characteristics of global relation modeling between any pixels. On this basis, we propose the first strong DETR-based baseline, remote sensing teacher, for UDA in remote sensing object detection. Specifically, the remote sensing teacher introduces an innovative learnable frequency-enhanced feature alignment (LFA) module. Within this module, we initially transform the features into frequency space to simplify the attention solver and effectively capture domain-specific information. Subsequently, the module significantly enhances the global feature representations of sparsely allocated objects by using a lightweight attention mechanism. Following this, the module incorporates learnable filters with a gated mechanism, enabling selective alignment of features in noisy backgrounds. In addition, the remote sensing teacher employs a self-adaptive pseudo-label assigner (SPA) that can automatically adjust the class-wise confidence threshold according to the model's learning status, thereby enabling the generation of high-quality pseudo-labels in scenarios with a long-tailed distribution. Leveraging these pseudo-labels further mitigates the domain bias of the detector by establishing alignment at the label level. Extensive experimental results demonstrate the superior performance and generalization capabilities of our proposed remote sensing teacher in multiple remote sensing adaptation scenarios. The Code is released at https://github.com/h751410234/RemoteSensingTeacher.**

***Index Terms*— Object detection, remote sensing imagery, unsupervised domain adaptation (UDA).**

Manuscript received 12 October 2023; revised 2 February 2024; accepted 7 March 2024. Date of publication 18 March 2024; date of current version 26 April 2024. This work was supported in part by the MYHT Program under Grant D040404, in part by the National Natural Science Foundation of China under Grant 62301046, and in part by the National Key Laboratory for Space-Born Intelligent Information Processing under Grant TJ-01-22-01. *(Jianhong Han and Wenjie Yang are co-first authors.) (Corresponding author: Yupei Wang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3378284

I. Introduction

OVER the past decade, deep learning-based object detection has made significant progress and has been extensively employed in the processing of remote sensing images. However, the majority of mainstream methods [1], [2], [3], [4] employ data-driven supervised architectures, which require a substantial amount of expensive labeled data. Moreover, the divergence in data distributions between the training and testing datasets, which is frequently encountered in remote sensing scenarios where two satellites employ sensors with different resolutions, bands, and so on, can lead to significant performance degradation for these methods.

To mitigate the aforementioned issues, many UDA techniques [5], [6], [7], [8] have been proposed to overcome the domain gap issue by leveraging unlabeled data from the target domain. The majority of existing unsupervised domain adaptive object detection approaches [7], [9], [10] are based on a convolutional two-stage detection architecture, such as faster regions with convolutional neural network (Faster RCNN) [3], and utilize elaborately designed feature alignment modules to enforce domain-invariant feature extraction.

With the remarkable success of DETR-based detectors, many researchers [11], [12], [13], [14], [15] have begun to explore the potential of the transformer to address the challenge of unsupervised domain adaption of object detection for remote sensing images. Different from convolutional neural network (CNN)-based detection architectures, DETR exploits a CNN backbone followed by an encoder–decoder structure-based transformer head, which allows it to learn effective feature representations [16]. Recent studies [17], [18], [19] have demonstrated that transformer is more adept at capturing global structural information during inference, and therefore have better generalization capabilities in target domain. On the other hand, DETR-based detectors can greatly simplify the design complexity of feature alignment modules by abandoning hand-crafted designs such as anchor generation and nonmaximum suppression [20].

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In remote sensing scenes, the sparsely allocated objects and inevitable background noises severely hinder the model from extracting domain-invariant features, resulting in suboptimal performance when directly applying unsupervised domain adaptive object detectors designed for natural scenes. In addition, in this research field, most researches focus on image classification and semantic segmentation, with relatively few studies [21], [22] dedicated to object detection. Therefore, it is highly meaningful to explore the performance of the transformer in addressing the unsupervised domain adaptation (UDA) challenge in remote sensing object detection.

In this article, we propose a remote sensing teacher, an effective and efficient unsupervised domain adaptive DETR for remote sensing object detection. Specifically, the remote sensing teacher introduces an innovative learnable frequency-enhanced feature alignment (LFA) module. With this alignment module, the detector accurately aligns cross-domain features in complicated remote sensing images. Moreover, the remote sensing teacher employs a self-adaptive pseudo-label assigner (SPA) that can automatically adjust the class-wise confidence threshold according to the model's learning status, thereby enabling the generation of high-quality pseudo-labels in scenarios with a long-tailed distribution.

The LFA module aims to help the detector effectively extract domain-invariant features in remote sensing scenes by achieving cross-domain feature alignment in the frequency space. Inspired by previous works [17], [23], [24], [25], [26], [27], we first develop an efficient frequency multihead cross-attention (FMC) mechanism within the LFA. This mechanism estimates scaled dot-product attention through element-wise product operations, rather than relying on matrix multiplication, thereby significantly reducing the computational complexity of the attention solver. By leveraging the proposed FMC, the high-level semantic features of the encoder can conditionally modulate the low-level local features from the backbone, enhancing the global feature representations of sparse objects. In addition, inspired by the property of frequency spectrum that domain-specific and domain-invariant information can be more effectively distinguished in the frequency space [17], we propose the learnable feed-forward network (LFFN). This network employs learnable filters with a gated mechanism to selectively align the enhanced features, enabling the detector to effectively extract domain-invariant features in remote sensing scenarios with noisy backgrounds.

The remote sensing teacher employs a self-training framework to establish cross-domain alignment at the label level by utilizing automatically generated high-quality pseudo labels for supervised learning, which can further mitigate the domain bias of the detector. Specifically, our framework consists of two distinct modules: a teacher model designed to process images from the target domain and a student model that handles images from both domains. The student model utilizes the proposed LFA to achieve cross-domain alignment at the feature level. The teacher detector is an ensemble model that employs exponential moving average (EMA) [28] to average the weights of the student detector. By feeding target domain images, the teacher model generates prediction results for the corresponding images. To obtain high-quality pseudo-labels in remote sensing scenes with a long-tailed category distribution, we introduce a SPA. This assigner adaptively adjusts the class-wise confidence threshold to filter prediction results based on the learning state of the teacher model, thereby avoiding the need for setting empirical hyperparameters and automatically generating high-quality pseudo-labels. Finally, the student network utilizes these pseudo-labels for supervised learning, further mitigating the domain bias of the detector.

The main contributions of this article are as follows.

1) We propose a remote sensing teacher, an effective and efficient DETR-based baseline for unsupervised domain adaptive remote sensing object detection. To the best of our knowledge, this is the first work that explores a DETR-based detector for unsupervised domain adaptive object detection for remote sensing images.
2) A LFA module is proposed to help the detector effectively extract domain-invariant features in remote sensing scenes by achieving cross-domain feature alignment in the frequency space.
3) We present a self-training framework to establish cross-domain alignment at the label level, aiming to further mitigate the domain bias of the detector. In this framework, we introduce an SPA that automatically generates high-quality pseudo-labels in remote sensing scenes with a long-tailed category distribution.

Extensive results demonstrate the superior performance and generalization capabilities of remote sensing teacher compared to state-of-the-art methods in multiple remote sensing adaptation scenarios. The adaptation from different optical satellites (xView [29] → DOTA [30]) exhibits a notable improvement of over 63.3% mean average precision (mAP). The adaptation between satellites and unmanned aerial vehicle (UAV) (UCAS-AOD [31] → CARPK [32], CARPK → UCAS-AOD) demonstrates remarkable enhancements in mAP exceeding 76.2% and 75.6%, respectively. These improvements are closely approaching the accuracy achieved through supervised learning on the target domain. The adaptation cross-modalities from optical to SAR images (HRRSD [33] → SSDD [34]) achieves 58.5% mAP, indicating the robust generalization capabilities of the remote sensing teacher.

II. Related Work

A. Unsupervised Domain Adaptive Object Detection

For object detection, the objective of UDA is to overcome the domain gap issue by leveraging unlabeled data from the target domain. Chen et al. [7] introduced the first UDA method for object detection, which has served as a foundation for subsequent advancements. The method includes two types of feature alignment: image-level and instance-level alignment, with the use of domain discriminators for adversarial training in each alignment module. Saito et al. [35] argued that larger shifts in the detection scenarios, such as the layout of the scene, the number of objects, etc., can hurt the performance of feature alignment. Therefore, they propose a weak global alignment with a strong local alignment to better achieve domain-invariant feature extraction. Zhu et al. [9] emphasized the importance of object local regions for detection

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and proposed an effective instance-level alignment module by utilizing the K-means algorithm to cluster candidate object regions.

With the remarkable success of the DETR, it has attracted widespread attention from researchers to explore its potential for UDA tasks. Wang et al. [12] proposed SFA to tackle the domain adaptive object detection via query-based feature alignment and token-wise feature alignment. Huang et al. [36] integrated adversarial feature alignment into detection transformers (DETRs) by implementing a novel adversarial token and a series of cross-attention layers. Zhang et al. [15] introduced a novel CNN-transformer blender (CTBlender) that fuses the CNN features and transformer features for effective feature alignment.

However, little research has been conducted on unsupervised domain adaptive object detection in remote sensing scenarios. FADA [21] introduced an adaptive feature alignment to alleviate the difficulty of aligning features between sparse foregrounds and complex backgrounds in remote sensing imagery. It is based on the single-stage detector RetinaNet [2]. Differently, we explore a DETR-based detector for unsupervised domain adaptive object detection in remote sensing, enabling effective feature alignment in complex scenes by transforming features to the frequency space.

B. Self-Training Framework

Self-training frameworks are widely used in semi-supervised object detection. These methods leverage a large amount of unlabeled data to generate pseudo labels, reducing the reliance on expensive annotated data during training. The unbiased teacher [37] proposed a student–teacher mutual learning pipeline to generate pseudo labels, and this framework has been extensively adopted in subsequent methods. Soft teacher [38] utilizes the scores of pseudo labels as weights in the loss calculation to generate more accurate pseudo labels. To relieve the label mismatch problem, LabelMatch [39] utilizes label distribution to dynamically determine the filtering threshold of different categories of pseudo labels. Efficient teacher [40] employs the self-training framework on a one-stage anchor-based detector (YOLOv5 [41]). Shen et al. [42] proposed the prospective consistency teacher (PCT), a novel teacher–student framework that effectively detects dense and multiscale geospatial objects while utilizing only a small amount of labeled data during training.

The aforementioned methods demonstrate great performance on CNN-based detectors, but their utilization on DETR-based detectors is largely neglected. MTTrans [14] attempts to transplant a mean teacher framework on DETR and generates pseudo labels using a fixed threshold. However, it is difficult for MTTrans to generate high-quality pseudo labels in remote sensing scenes due to the significant distribution differences among categories. Differently, we employ a self-training framework in the DETR architecture to achieve cross-domain alignment at the label level. In this framework, our proposed self-adapting pseudo-label assigner can improve the quality of pseudo-labels in remote sensing scenarios with long-tailed distribution and avoid the need for empirical hyperparameter settings.

III. Remote Sensing Teacher

Remote sensing teacher is an effective and efficient DETR-based detector for UDA object detection in remote sensing imagery. The framework is based on deformable-DETR [43] and the student–teacher mutual learning approach. The overview of our method is shown in Fig. 1. First, we present the preliminaries of the DETR. Subsequently, we detail the framework of the remote sensing teacher and our key components. Our proposed LFA module helps the detector accurately align cross-domain features in complex remote sensing images with sparse foregrounds. The SPA can automatically adjust the class-wise confidence threshold according to the model's learning status, thereby enabling the generation of high-quality pseudo-labels in scenarios with a long-tailed distribution.

A. Preliminaries of DETR

DETR [4] is a transformer-based end-to-end object detector that eschews traditional hand-designed components such as anchor design and NMS. Specifically, DETR integrates a CNN backbone [44] for feature extraction, which is then utilized by an encoder–decoder transformer structure and a feed-forward network (FFN) for final detection prediction. For example, when processing an image, the CNN backbone first generates a single-scale feature map. This feature map is then reshaped into a vector, feeding into the DETR's encoder–decoder. The encoder–decoder of DETR adheres to the transformer's standard architecture [45] and includes several multihead attention modules. The encoder primarily comprises several multihead self-attention modules to enhance the input feature map. The decoder is stacked with several multihead self-attention and cross-attention modules, transforming the input object queries into output embeddings. Finally, using these output embeddings, the final prediction is computed by a three-layer perceptron with rectified linear unit (ReLU) activation function.

Deformable-DETR [43] was proposed to enhance the convergence speed of the conventional DETR. This innovative variant refines the standard attention mechanism by incorporating deformable attention mechanisms. In addition, deformable-DETR aggregates multiscale features for prediction, as shown in Fig. 2(a). Specifically, the backbone first produces multiscale CNN features $F^l (l = 1, 2, 3, 4)$, which are then processed by the transformer encoder to generate Transformer features $P^l (l = 1, 2, 3, 4)$. Ultimately, similar to the DETR, the decoder and the FFN are utilized to compute the final prediction results.

B. Framework Overview

As shown in Fig. 1, our remote sensing teacher is an iterative teacher–student learning framework consisting of two models with identical architecture. Each model includes a base detector and an elaborate adversarial training plugin. We adopt the deformable-DETR as the base detector, and the adversarial training plugin includes the proposed LFA module and a discriminator. We follow [46] to update the parameters of the teacher network from the student network via

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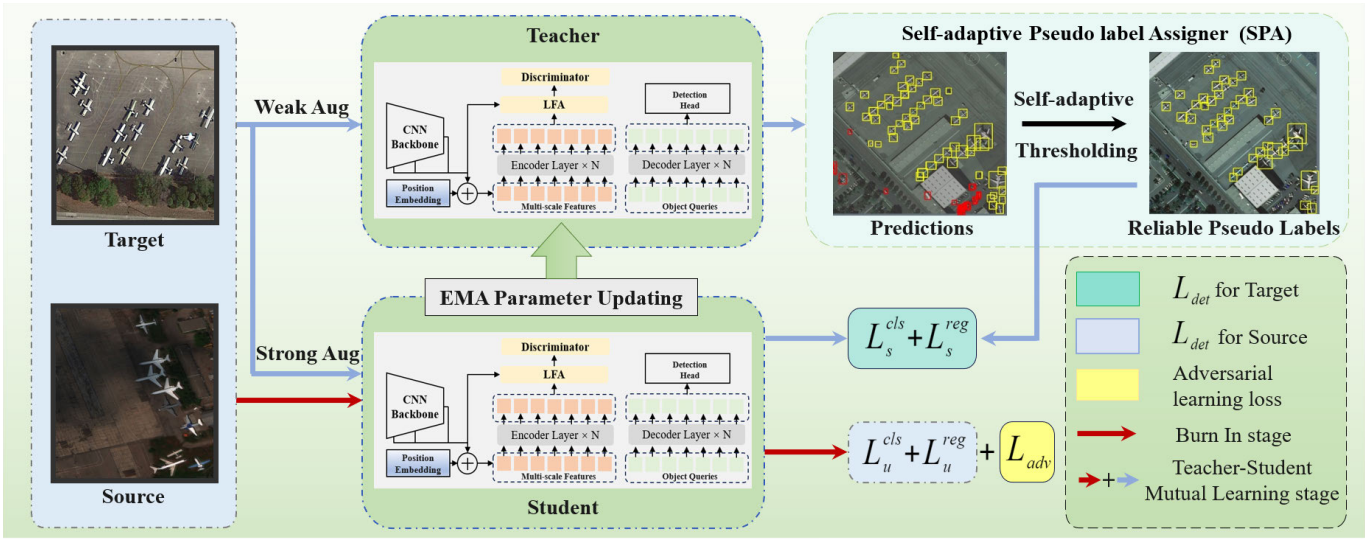


Fig. 1. Framework of the remote sensing teacher consists of a student model, which is the actual task model, and a temporally ensembled version of the student model called the teacher model. The student model is comprised of a base detector and the proposed LFA module. This module helps the detector in effectively extracting domain-invariant features in remote sensing scenes by achieving cross-domain feature alignment in the frequency space. The teacher model is updated by the EMA of the student model and generates high-quality pseudo labels for the target domain by utilizing the newly proposed SPA. The remote sensing teacher leverages these pseudo-labels to further mitigate the domain bias of the detector by establishing alignment at the label level. We divide the training process into two stages to enable rapid convergence of the network without reducing accuracy. In the burn-in stage, we employ only adversarial training to achieve cross-domain feature alignment. In the teacher–student mutual learning stage, the unlabeled data in the target domain is fed into the teacher model to generate pseudo labels for supervised learning.

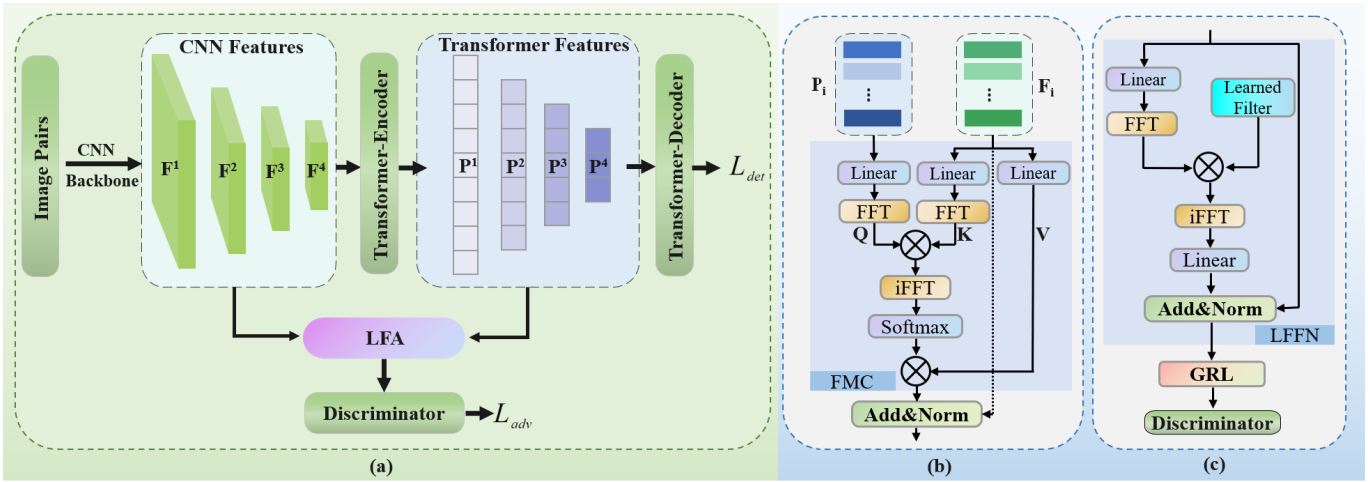


Fig. 2. Details of (a) proposed DETR-based detector that contains LFA module for cross-domain feature alignment, (b) proposed FMC mechanism, and (c) proposed LFFN.

EMA [28], as follows:

$$\theta_t^i \leftarrow \alpha\theta_t^{i-1} + (1-\alpha)\theta_s^i \tag{1}$$

where θ_t and θ_s are the parameters of the teacher and student networks, respectively, and i denotes the training step. α is the hyper-parameter to determine the speed of parameter transmission, which is normally close to 1.

The self-training framework always faces the challenge of achieving stability and high efficiency during training. In order to enable the rapid convergence of the network without reducing accuracy, we divide the training process into two stages: the burn-in stage and the teacher–student mutual learning stage.

In the burn-in stage, we only employ adversarial training for cross-domain feature alignment. Specifically, the pipeline is illustrated in Fig. 2(a). The network input is a batch of image pairs, consisting of an equal number of images from the source and target domains. Given an image pair, we first use a backbone Resnet [44] to extract multiscale features $F^l(l = 1, 2, 3, 4)$. These features are then enhanced by the transformer encoder, resulting in transformer features $P^l(l = 1, 2, 3, 4)$. For supervised learning, the decoder only uses the transformer features generated from the source domain to predict the detection results because no pseudo-labels are generated. Similar to deformable-DETR, the detection loss L_{det} is computed by using several loss functions to measure the discrepancy between the predicted and the corresponding

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ground truth. For unsupervised learning, we propose the LFA to achieve more effective cross-domain feature alignment. Our LFA utilizes the FMC mechanism to efficiently enhance the global feature representations of sparsely allocated objects. Furthermore, selectively align feature extraction within the network by utilizing learnable filters with a gated mechanism. More details of LFA are available in Section III-C. Finally, the output of LFA is fed into a simple MLP-based discriminator D to calculate the adversarial loss L_{adv}. By optimizing L_{adv}, the backbone and the transformer learn to precisely extract domain-invariant features. Through optimizing L_{det}, the decoder can utilize transformer features for predicting object localization and categories. Further elaboration on the loss function can be found in Section III-E.

In the teacher–student mutual learning stage, unsupervised learning is identical to the burn-in stage. Differently, the unlabeled data in the target domain is fed into the teacher model to generate pseudo labels for supervised learning. Beyond the existing self-training method [14], [37], [38], [46], [47], [48], we propose a SPA that automatically generates high-quality pseudo-labels for remote sensing scenes with long-tailed distribution. Leveraging these pseudo-labels further mitigates the domain bias of the student model by establishing alignment at the label level. The SPA will be presented in Section III-D.

C. LFA Module

One critical component of the remote sensing teacher is the LFA module, which serves two key purposes: employing a concise adversarial learning architecture and achieving accurate feature alignment by transforming features into the frequency space. The presented LFA adopts multiscale CNN features along with corresponding multiscale transformer features as inputs and utilizes the proposed FMC mechanism to enhance the global feature representations of objects. Subsequently, the LFFN selectively aligned the prominent features by utilizing learnable filters with a gated mechanism.

1) FMC Mechanism: Existing adversarial training methods [5], [6], [7], [8], [13], [36] achieve independent adversarial training for different parts of the detector, such as the backbone, neck, and head, by using multiple discriminators. However, these approaches not only lead to a lack of coordinated feature alignment among the detector's components but also significantly increase the computational burden during network training. To address this issue, our proposed FMC mechanism achieves information communication between the encoder and the backbone and conducts adversarial learning using only a single discriminator. Moreover, by utilizing element-wise attention in the frequency space, the high-level semantic features of the encoder can conditionally modulate the low-level local features from the backbone, enhancing the global feature representations of sparse objects in the low-level features with fewer computations. The detailed network architecture of the proposed FMC is shown in Fig. 2(b).

Given a transformer features $P^l(l = 1, 2, 3, 4)$ and a backbone CNN features $F^l(l = 1, 2, 3, 4)$ (where l represents different hierarchical levels of feature maps). First, we calculate the query Q, key K, and value V by applying linear transformation matrices W_q, W_k, and W_v, respectively, as shown in the following equation:

$$Q = W_q F, \quad K = W_k P, \quad V = W_v F. \tag{2}$$

Then, inspired by the convolution theorem, we compute the attention weight in the frequency space. By transforming the features into the frequency space, we efficiently estimate the attention map using element-wise multiplication, thereby avoiding the computational burden of the dot-product operation. Specifically, we transform the Q and K to the frequency space by applying the fast Fourier transform (FFT) and obtain the feature correlation coefficients through element-wise multiplication. After performing the inverse FFT (iFFT) and applying the softmax function, we derive the ultimate attention weight A. The entire process is summarized as follows:

$$A = \text{Softmax}(\mathcal{F}^{-1}(\mathcal{F}Q \times \overline{\mathcal{F}K})) \tag{3}$$

where $\mathcal{F}(\cdot)$ denotes the FFT, $\overline{\mathcal{F}(\cdot)}$ represents the complex conjugate of the FFT, $\mathcal{F}^{-1}(\cdot)$ denotes the iFFT, and $\times$ represents element-wise multiplication.

Finally, the refined output feature map $\hat{F}$, can be obtained by

$$\hat{F} = \mathcal{L}(A \times V + V) \tag{4}$$

where a layer norm $\mathcal{L}(\cdot)$ is used to normalize $\hat{F}$.

Similar to spatial attention mechanisms, we also employ channel dimension compression to achieve multihead attention. In this work, we utilize $h = 8$ parallel attention heads, resulting in a reduction of the channel size for Q, K, and V to $d/h = 256/8 = 64$.

2) Learnable FFN: The FFN represents the only nonlinear component in the feature alignment module, thus playing a crucial role in feature processing. Numerous studies [17], [23], [24], [25], [49] have attempted to develop a more efficient variant of FFN to achieve better performance for the task. Inspired by the property of frequency spectrum that domain-specific and domain-invariant information can be more effectively distinguished in the frequency space [17], [50], we propose the LFFN. Specifically, our proposed LFFN is composed of a sequence of stacked linear layers and a frequency-space learnable filter. We utilize this filter to selectively align the prominent features. The formulation of the proposed LFFN can be expressed as

$$\begin{aligned} X_1 &= \mathcal{F}^{-1}(\mathcal{F}(\text{Linear}(X)) \times W) \\ X_{\text{out}} &= \text{Linear}(\text{GELU}(X1)) + X \end{aligned} \tag{5}$$

where Linear denotes the linear layer, W represents the frequency-space learnable filter, and GELU denotes the Gaussian error linear unit activation function [51]. The detailed network architecture of the proposed LFFN is shown in Fig. 2(c).

D. Self-Adaption Pseudo Label Assigner

Our objective is to further mitigate the domain bias of the detector by establishing alignment at the label level through the utilization of pseudo-labels. Meanwhile, the pseudo-label

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assigner serves as a simple filter to eliminate prediction results below a predefined threshold. Predictions with scores below the threshold are disregarded, while those with scores above the threshold are considered reliable pseudo-labels. However, existing methods [14], [37], [47], [52], [53] that use a fixed threshold are suboptimal and fail to adapt to the long-tailed distribution observed in remote sensing scenes. First, as the iterations increase, the varying predicted status of the model makes it challenging to employ a fixed threshold throughout all training stages. This leads to unstable training and the generation of low-quality pseudo-labels. Moreover, the model exhibits varying capabilities for different classes, making it meaningful to establish an adaptive threshold for each category.

Inspired by Wang et al. [54], we have devised an SPA that leverages model predictions to autonomously establish and dynamically adjust confidence thresholds during the training. Specifically, the proposed SPA introduces two types of thresholds: the global threshold and the local threshold. By using the local threshold to modulate the global threshold, our method enables fine-tuned adjustments of confidence thresholds for each class.

1) Global Threshold: Throughout the entire self-training process, the global threshold reflects the overall learning state of the model. Hence, it should maintain stable variations that align with the model's capabilities, ensuring the proper filtering of incorrect pseudo-labels. We initialize the global threshold τ_t as 0.5 [14]. For each training iteration, we compute the average confidence score M_s across all detection results for a batch of unlabeled data. To update the global threshold, we employ the EMA method based on the M_s. The global threshold τ_t is defined and adjusted as follows:

$$\tau_t = \begin{cases} 0.5, & \text{if } t = 0 \\ \lambda \tau_{t-1} + (1-\lambda)\dfrac{1}{B}\displaystyle\sum_{b=1}^{B} M_s, & \text{otherwise} \end{cases} \tag{6}$$

where $\lambda \in (0, 1)$ is the momentum decay of EMA. B represents the number of batch sizes for the unlabeled data.

2) Local Threshold: The local threshold adjusts the global threshold according to the learning status of different categories, resulting in class-specific thresholds. Similar to the global threshold, we first compute the expectation of the detection results for each class c in a batch of unlabeled data. Subsequently, we utilize the EMA to estimate the local threshold, as illustrated below

$$\tilde{p}_t(c) = \begin{cases} 0.5, & \text{if } t = 0 \\ \lambda \tilde{p}_{t-1}(c) + (1-\lambda)\dfrac{1}{B}\displaystyle\sum_{b=1}^{B} M_c(c), & \text{otherwise} \end{cases} \tag{7}$$

where $M_c(c)$ denotes the expectation the detection results for class c, $\tilde{p}_t(c)$ represents local threshold for class c. The final self-adaptive threshold $\tau_c(c)$ can be obtained by modulating the global threshold using the local threshold, as shown in the following equation:

$$\tau_c(c) = \text{MaxNorm}\big(\tilde{p}_t(c)\big) \cdot \tau_t = \frac{\tilde{p}_t(c)}{\max\{\tilde{p}_t(c) : c \in [C]\}} \cdot \tau_t \tag{8}$$

where MaxNorm is the maximum normalization. $\tilde{p}_t = [\tilde{p}_t(1), \tilde{p}_t(2), \ldots, \tilde{p}_t(c)]$ is the list containing all $\tilde{p}_t(c)$.

E. Network Training

The remote sensing teacher is trained with two loss functions: the supervised detection loss L_{det} and the adversarial learning loss L_{adv}. As illustrated in Fig. 1, the adversarial learning loss L_{adv} remains consistent across both two training stages and can be defined as follows:

$$L_{\text{adv}} = -\sum_{N}\big[D \log p_{(N)} + (1-D)\log\big(1 - p_{(N)}\big)\big] \tag{9}$$

where $p_{(N)}$ represents the output of the domain classifier. $D = 0$ for labeled data and $D = 1$ for unlabeled data. The gradient reverse layer (GRL) [55] is employed to process the input features of the domain discriminator, enabling adversarial training through gradient reversal.

In the burn-in stage, the supervised detection loss is exclusively trained using data from the source domain. The training objective of the Remote Sensing Teacher during this stage can be defined as follows:

$$\begin{aligned} L &= L_{\text{det}}^{\text{sup}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \\ &= \mathcal{L}_{\text{det}}\left(P_{\text{src}}, Y_{\text{src}}\right) + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \end{aligned} \tag{10}$$

where P_{src} represents the predicted bounding box for source data, Y_{src} denotes the ground truth, and $L_{\text{det}}^{\text{sup}}$ denotes the supervised object detection loss, which remains consistent with deformable-DETR. The L_{adv} is defined in (10), λ_{adv} is used to balance the supervised detection loss and the adversarial learning loss.

Subsequently, in the teacher–student mutual learning stage, additional unsupervised data is incorporated, and reliable pseudo-labels generated through detection are employed for supervised training. Therefore, the training objective of the remote sensing teacher during this stage is defined as follows:

$$\begin{aligned} L &= L_{\text{det}}^{\text{sup}} + \lambda_{\text{unsup}} L_{\text{det}}^{\text{unsup}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \\ &= \mathcal{L}_{\text{det}}\left(P_{\text{src}}, Y_{\text{src}}\right) + \lambda_{\text{unsup}} \mathcal{L}_{\text{det}}\left(P_{\text{tgt}}, Y_{\text{tgt}}\right) + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \end{aligned} \tag{11}$$

where P_{tgt} denotes the predicted bounding box for the target data, Y_{src} is the generated pseudo labels, $L_{\text{det}}^{\text{unsup}}$ represents the unsupervised object detection loss, and λ_{unsup} denotes the balancing weights for the corresponding learning loss function.

IV. Experiments

This section presents experimentation including remote sensing datasets, evaluation metrics, implementation details, ablation studies, comparisons with the state-of-the-art, visualization, and analysis. Further elaboration on these aspects is provided below.

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TABLE I
DATASET OVERVIEW FOR DIFFERENT DOMAIN ADAPTATION SCENARIOS

domain adaptation scenarios	dataset	sensor	class	GSD	image size	images
between different optical satellites	DOTA xView	GoogleEarth+ WorldView-3	Pla+Ship+ST	$0.1 \sim 4.5$ 0.3	$800 \times 800 \sim 4k \times 4k$ $2.5k \times 2.5k \sim 5k \times 3k$	792 216
between satellite and UAV	UCAS-AOD CARPK	GoogleEarth UAV	Car	— $0.01 \sim 0.04$	1280×659 1280×720	510 1448
from optical to SAR images	HRRSD SSDD	GoogleEarth RadarSat-2+	Ship	$0.15 \sim 1.2$ $1 \sim 15$	$1k \times 1k \sim 2k \times 2k$ $300 \times 300 \sim 500 \times 500$	2165 1160

- GoogleEarth+: Google Earth, Satelite JL-1, GF-2
- RadarSat-2+: RadarSat-2+, TerraSAR-X, Sentinel-1
- Pla+Ship+ST: Plane, Ship, Storage-tank

A. Datasets

In this study, we evaluated the effectiveness and generalization capabilities of the remote sensing teacher under three widely adopted domain adaptation scenarios with six public datasets. The source and target datasets utilized in our devised scenarios are openly accessible. The purpose of these experimentations is to establish a standardized benchmark for cross-domain object detection in the field of remote sensing. First, following [21], we evaluated the cross-domain knowledge transfer of the detector between different satellites (xView [29] → DOTA [30]). Then, we demonstrate the proposed effectiveness between satellite and UAV (UCAS-AOD [31] → CARPK [32], CARPK → UCAS-AOD). Moreover, to validate the generalization ability of our approach, we follow [22] and evaluate the performance of our method cross modalities from optical to SAR images (HRRSD [33] → SSDD [34]). These datasets are compared in Table I and introduced in the following paragraphs.

1) DOTA: The DOTA dataset, published by Wuhan University, is the largest publicly available optical remote sensing image dataset. It is primarily sourced from Google Earth and includes 15 categories, such as airplanes, ships, vehicles, and more. Through elaborate design to represent the data distributions of real-world remote sensing images, the dataset comprises 188 282 instances across 2806 images. To align with xView remote sensing datasets in terms of categories, we selected three categories for our experiments: airplane, ship, and storage-tank. The remote sensing teacher was trained on the training set and evaluated on the validation set, both of which were officially partitioned. We crop the original images into 800 × 800 patches, with an overlapped stride of 100 pixels, as input for the network.

2) xView: Following [21], this dataset is used alongside DOTA to evaluate the method's knowledge transfer across different satellites. Different from DOTA, xView is captured by the WorldView-3 satellite. It contains over one million annotated bounding boxes from diverse complex scenes worldwide. Due to the fine-grained labeling in xView, we merged subcategories into the superior xView category to align with DOTA's category setting. For example, the motorboat, sailboat, and fishing vessel were merged into the ship category. Our experiments used officially partitioned training sets and focused on the plane, ship, and storage-tank categories. We crop the original images into 800 × 800 patches, with an overlapped stride of 100 pixels, as input for the network.

3) UCAS-AOD: The dataset was released by the University of Chinese Academy of Sciences (UCAS) in 2014 and obtained from Google Earth. It includes carefully selected images to ensure a uniform distribution of object orientations. For the experiments, we focused on the car data, consisting of 510 images and 7114 vehicles. Since the UCAS-AOD image size is 1280 × 659 pixels, there was no need for image cropping. In addition, the dataset was not uniformly partitioned by the official source. To ensure result reproducibility, we selected images with filenames ending in digits 1, 5, and 9 as the test set, while the remaining images were used for training. This maintained a training-to-test set ratio of 7:3.

4) CARPK: The public CARPK dataset is captured by a UAV in parking lots. Each image has a size of 1280 × 720 pixels and exclusively contains vehicles of a single category. To inspect the effectiveness of our remote sensing teacher, we paired this dataset with UCAS-AOD to achieve domain adaptation between satellite and UAV imagery.

5) High-Resolution Remote Sensing Detection (HRRSD): The HRRSD dataset was released by the University of Chinese Academy of Sciences in 2019. It comprises thousands of images sourced from Google Earth and Baidu Maps, with spatial resolutions ranging from 0.15 to 1.2 m. For our experiments, we selected ship data, including 2165 images containing a total of 3975 ships. To accommodate variations in image sizes, we resized the input images to have a minimum short edge length of 600 pixels, and all samples were utilized for training.

6) SAR Ship Detection Dataset (SSDD): The SSDD is a publicly available dataset created by Li et al. [34]. It comprises a total of 1160 images and includes 2456 ships, capturing diverse variations in factors such as sea state, resolution, and sensors. Following [22], this dataset is paired with the HRRSD for the purpose of evaluating the cross-modalities domain adaptation capability of methods from optical images to SAR images. In the experiments, we resized the input images to have a minimum short edge length of 600 pixels. Subsequently, training was conducted on the officially partitioned training set, and results were evaluated using the test set.

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B. Evaluation Metric

In this article, we employ standard evaluation metrics for detection tasks to evaluate the performance of models. Specifically, average precision (AP) is utilized, which represents the area under the precision–recall curve and combines precision and recall. The expression is

$$\mathrm{AP} = \int_0^1 P(R)dR \tag{12}$$

where P represents the precision rate and R is the recall rate. We ultimately used mAP to evaluate the performance of the detector, where mAP represents the average AP value of multiple categories, as follows:

$$\mathrm{mAP} = \frac{1}{c}\sum_{i=0}^{c} \mathrm{AP}_i \tag{13}$$

where c represents the number of categories.

C. Implementation Details

We adopt deformable-DETR as the base detector. Our method compares with both CNN-based and transformer-based domain adaptive detection methods. To ensure a fair comparison, all methods aim to use ResNet-50 (pretrained on ImageNet [56]) as the backbone network whenever possible. In all experiments, our model is trained for a maximum of 60 epochs, with the first 50 epochs designated as the burn-in stage, followed by 10 epochs dedicated to the teacher–student mutual learning stage. In the burn-in stage, we train our models using Adam optimizer [57] with a base learning rate of 2×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and the learning rate is decayed at the 40th epoch by a factor of 0.1. In the teacher–student mutual learning stage, we train our models using Adam optimizer with a base learning rate of 2×10^{-4}. For all scenarios, we set the weight factor λ_{adv} in (11) and (12) to 0.1 and λ_{unsup} in (12) to 3.0. The smoothing hyper-parameter in the EMA is set to 0.999. We conduct each experiment on 4 NVIDIA A6000 with 48G memory per GPU.

D. Ablation Studies

To effectively analyze the contribution of each component in our proposed Remote sensing teacher, we conducted ablation experiments by replacing or removing parts of the components. Our remote sensing teacher consists of adversarial learning and a self-training framework for cross-domain alignment. We first study the contribution of the LFA module. Next, we explore the benefits of a self-training framework for DETR-based detectors in UDA and demonstrate the effectiveness of the proposed SPA. Table II shows experimental results on the validation data of DOTA under the adaptation scenario from xView to the DOTA task.

The results are shown in Table II. Training deformable-DETR solely on the source domain data proves challenging in achieving favorable results due to domain shifts. "Direct-align" refers to the utilization of adversarial learning to align the output features of the CNN backbone, which leads to a 3.6% improvement in mAP. In comparison to "Direct-align," our proposed LFA demonstrates superior cross-domain alignment at the feature level, achieving a 7.1% improvement in mAP. Furthermore, we explore the effectiveness of a self-training framework for DETR-based detectors in UDA. We first observed that transplanting the mean teacher [58] to deformable-DETR with a fixed pseudo-label generation threshold of 0.5 resulted in a further 3.8% improvement in mAP. By utilizing the proposed SPA for adaptive pseudo-label generation, we achieve a significant improvement of 1.4% over the fixed threshold version. Ultimately, our remote sensing teacher outperforms the base detector with a remarkable 12.3% increase in mAP, resulting in a final performance of 63.3%.

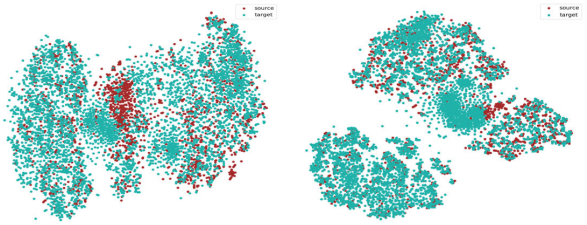
The hyperparameter λ_{adv} functions as the weighting coefficient for our proposed LFA module within (11) and (12), balancing the adversarial learning loss and detection loss. Following [59], we set λ_{adv} to 0.01, 0.05, and 0.1 to investigate its impact on our feature alignment module under the adaptation scenario from xView to the DOTA task. The specific results are presented in Table III. It can be observed that our proposed LFA module effectively helps the detector in extracting domain-invariant features, thereby improving the cross-domain detection performance under various weight settings. On the other hand, increasing the weights can lead to slightly improved performance, attributable to the enhanced role of the LFA module during training. In this article, we set the weight to 0.1 to maintain consistency with existing methods [7], [12], [36], [60] for a fair comparison.

TABLE II
ABLATION STUDY OF REMOTE SENSING TEACHER

adversarial learning		self-training		mAP_{50}
Direct-align	LFA	Fixed threshold	SPA	
				51.0
√				54.6
	√			58.1
	√	√		61.9
	√		√	**63.3**

TABLE III
ABLATION STUDY OF THE HYPERPARAMETER λ_{ADV}

Method	λ_{adv}	mAP_{50}
Deformable-DETR	—	51.0
Deformable-DETR + LFA	0.01	56.1
	0.05	57.4
	0.1	58.1



(a) (b)

Fig. 3. Feature visualizations. Our method aligns the domain shift well compared to the baseline method. (a) Base detector. (b) Remote sensing teacher.

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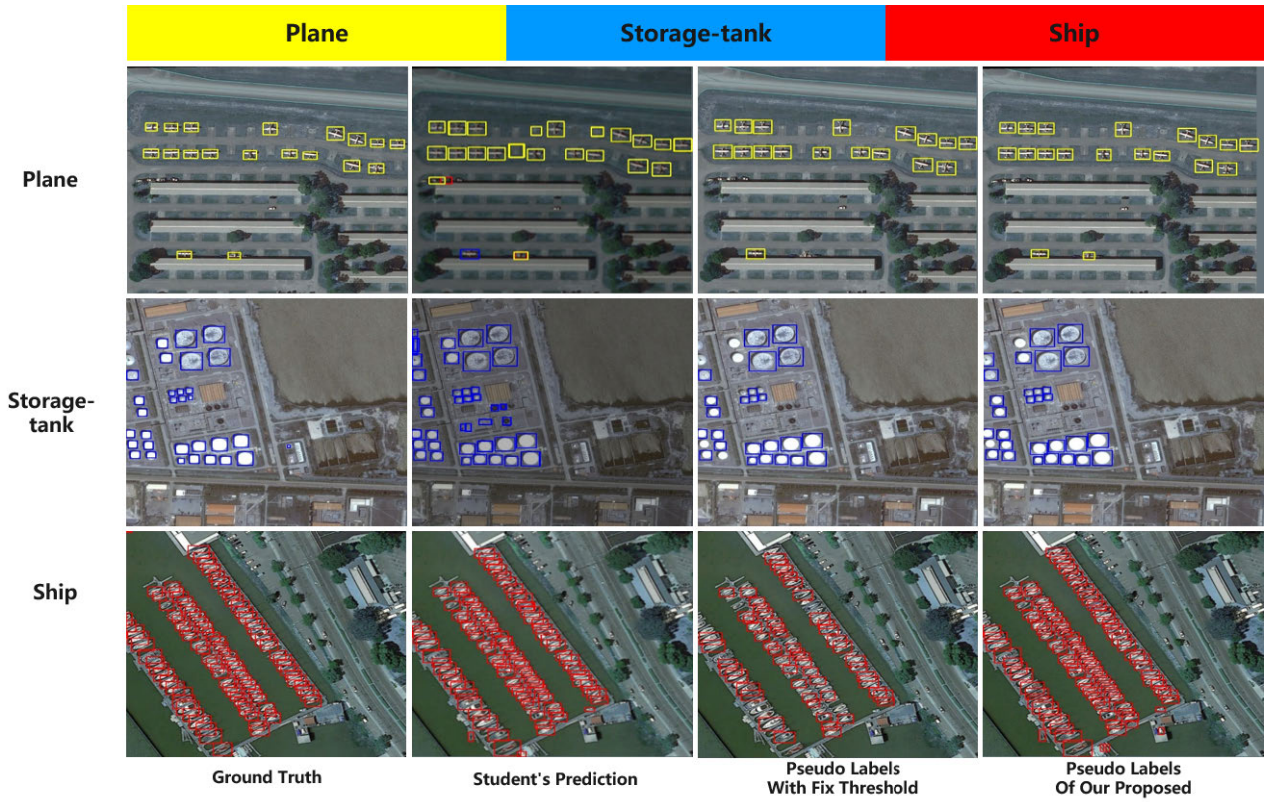


Fig. 4. Visualization includes pseudo labels generated by the remote sensing teacher, predictions of the student model, the ground truth annotations, and the remote sensing teacher's fixed threshold version set at 0.5. As seen in the visualization results, our proposed method can generate pseudo labels of higher quality compared to the fixed threshold version. However, there remains a gap in quality between the pseudo-labels and the ground truth annotations, especially in dense or small target scenarios, such as ships.

TABLE IV
ABLATION STUDIES OF THRESHOLD VALUE

Method	Threshold Value	mAP_{50}
Deformable-DETR + LFA	—	58.1
Deformable-DETR + LFA + Self-training	0.4	62.1
	0.5	61.9
	0.6	60.7
	0.7	61.4
	SPA(Ours)	**63.3**

We further analyze the hyperparameter related to the confidence threshold value for generating pseudo labels, aiming to reinforce the efficacy of our proposed SPA. Following the methodology of Xu et al. [40], we adjusted the threshold values to 0.4, 0.5, 0.6, and 0.7 to examine their impact on performance under the adaptation scenario from xView to the DOTA task. The detailed results are presented in Table IV. Altering the set threshold value significantly affects the detection results, but there remains a gap between a fixed threshold value and our SPA method. Our results demonstrate that the SPA can automatically adjust the class-wise confidence threshold in accordance with the model's learning status, enabling the generation of high-quality pseudo-labels while mitigating the negative impacts of manual settings.

E. Comparisons With the State-of-the-Art

To showcase the effectiveness and generalization capability of the remote sensing teacher, we evaluate our proposal across three elaborate domain adaptation scenarios: 1) between different optical satellites; 2) between satellite and UAV; and 3) cross-modalities from optical to SAR images. In each scenario, we compare remote sensing teacher with several state-of-the-art unsupervised domain adaptive methods, including both CNN-based and transformer-based detectors. Since the existing advanced works focus on natural scenes, we reimplemented them in the remote sensing scenarios with their official default settings. In addition, both the source and target datasets used in our designed scenarios are publicly available. We aim for this report to serve as a benchmark for domain adaptive object detection in remote sensing, thereby contributing to further research in this domain.

1) Adaptation Between Different Optical Satellites: We first investigate the adaptation between different optical satellites on the task xView → DOTA. As shown in Table V, the remote sensing teacher achieves the best mAP of 63.3%, significantly outperforms the baseline deformable-DETR and surpassing the state-of-the-art method by 5.0% in mAP. It demonstrates the effectiveness of our method in typical remote sensing cross-domain scenarios.

2) Adaptation Between Satellite and UAV: Table VI shows experiments of adaptation between satellite and UAV. First, on the task UCAS-AOD → CARPK. It can be observed that the remote sensing teacher achieves the highest accuracy with an mAP of 76.2%. On the other hand, we also conducted experiments of adaptation from UAV to satellite scenario on the task CARPK → UCAS-AOD. Our remote sensing

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TABLE V
EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN DIFFERENT OPTICAL SATELLITES: XVIEW → DOTA

xView → DOTA								
Method	Type	Backbone	Source	Target	Plane	Ship	Storage-tank	mAP_{50}
Deformable-DETR(ICLR'21) [43]	Transformer	resnet-50	√		64.8	55.5	32.6	51.0
				√	93.5	74.1	76.2	81.3
DAF(CVPR'18) [7]	CNN	resnet-50	√		57.3	34.3	36.4	42.7
SWF(CVPR'19) [35]	CNN	resnet-50	√		48.1	33.9	34.3	38.7
EPM(ECCV'20) [60]	CNN	resnet-101	√		60.5	59.8	43.9	54.8
SFA(ACM MM'21) [12]	Transformer	resnet-50	√		65.2	57.1	38.9	53.7
AQT(IJCAI'22) [36]	Transformer	resnet-50	√		68.2	**59.9**	46.8	58.3
FADA(TGRS'22) [21]	CNN	resnet-50	√		68.0	43.4	46.2	52.5
Remote Sensing Teacher	Transformer	resnet-50	√		**75.7**	59.7	**54.7**	**63.3**

TABLE VI
EXPERIMENTAL RESULTS (%) OF THE SCENARIO SATELLITE TO UAV: UCAS-AOD → CARPK AND UAV TO SATELLITE: CARPK → UCAS-AOD

UCAS-AOD → CARPK					
Method	Type	Backbone	source	target	mAP_{50}
Deformable-DETR(ICLR'21) [43]	Transformer	resnet-50	√		61.2
				√	80.9
DAF(CVPR'18) [7]	CNN	resnet-50	√		68.4
SWF(CVPR'19) [35]	CNN	resnet-50	√		58.3
EPM(ECCV'20) [60]	CNN	resnet-101	√		71.2
SFA(ACM MM'21) [12]	Transformer	resnet-50	√		49.8
AQT(IJCAI'22) [36]	Transformer	resnet-50	√		71.0
Remote Sensing Teacher	Transformer	resnet-50	√		**76.2**
CARPK → UCAS-AOD					
method	Type	Backbone	source	target	mAP_{50}
Deformable-DETR(ICLR'21) [43]	Transformer	resnet-50	√		58.1
				√	79.1
DAF(CVPR'18) [7]	CNN	resnet-50	√		53.7
SWF(CVPR'19) [35]	CNN	resnet-50	√		57.1
EPM(ECCV'20) [60]	CNN	resnet-101	√		65.0
SFA(ACM MM'21) [12]	Transformer	resnet-50	√		66.1
AQT(IJCAI'22) [36]	Transformer	resnet-50	√		70.8
Remote Sensing Teacher	Transformer	resnet-50	√		**75.6**

TABLE VII
EXPERIMENTAL RESULTS (%) OF THE SCENARIO FROM OPTICAL TO SAR IMAGES: HRRSD → SSDD

HRRSD → SSDD					
Method	Type	Backbone	source	target	mAP_{50}
Deformable-DETR(ICLR'21) [43]	Transformer	resnet-50	√		27.8
				√	93.6
DAF(CVPR'18) [7]	CNN	resnet-50	√		47.0
SWF(CVPR'19) [35]	CNN	resnet-50	√		52.5
EPM(ECCV'20) [60]	CNN	resnet-101	√		54.9
SFA(ACM MM'21) [12]	Transformer	resnet-50	√		34.3
AQT(IJCAI'22) [36]	Transformer	resnet-50	√		39.9
Remote Sensing Teacher	Transformer	resnet-50	√		**58.5**

teacher significantly improved the baseline accuracy, achieving a mAP of 75.6%, closely approximating the accuracy achieved through supervised learning using target domain data. The experiments demonstrate the powerful capability of the remote sensing teacher in addressing straightforward cross-domain detection tasks.

3) Adaptation From Optical to SAR Images: We evaluate the cross-modalities adaptation from optical to SAR images on the task HRRSD → SSDD. Table VII shows experimental results, where the remote sensing teacher achieves the best mAP of 58.5%. In this experiment, we observed that the performance of the DETR-based detector is inferior to that of

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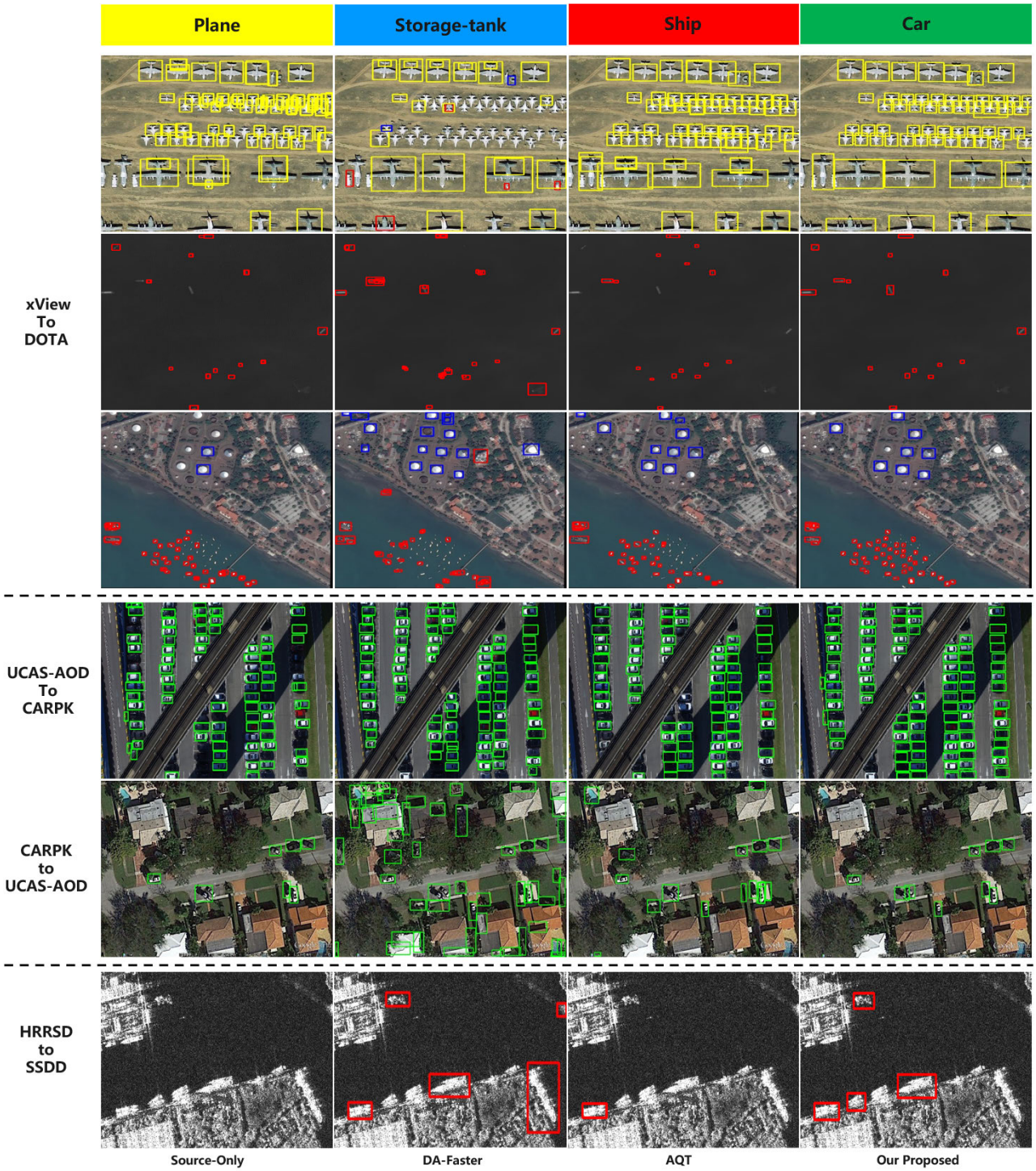


Fig. 5. Our detection results compared with the state-of-the-art methods. Different categories are marked with different colors. The confidence threshold for visualization is set to 0.2. "Source-only" represents the base detector that uses only source domain data for training. Other descriptions represent their respective methods.

the CNN-based detector in cross-modality domain adaptation tasks. In contrast, our proposed effectively addresses this issue by establishing alignment at the label level to mitigate the domain bias of the detector. The outcome of this experiment compellingly demonstrates the generalization capability of remote sensing teacher.

F. Visualization and Analysis

1) Feature Visualization: By utilizing the t-distributed stochastic neighbor embedding (t-SNE) method [61], we visualized the features extracted from the backbone on the task xView → DOTA, as shown in Fig. 3. Evidently, a base detector trained solely on the source domain exhibits significant cross-domain gaps, making it easy to distinguish different domain features. In contrast, our proposed remote sensing teacher effectively mixtures different domains, enabling the detector to extract domain-invariant features and improve cross-domain performance.

2) Pseudo Label Visualization: We show visualization results of the pseudo labels generated by the remote sensing teacher on the xView → DOTA task. These results are compared with the predictions of the student model, the ground

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truth, and the fixed threshold version of the remote sensing teacher, as shown in Fig. 4. It is worth noting that the input images of the student model undergo strong augmentation during training, while the input images of the teacher model are subject to weak augmentation. Consequently, the pseudo labels generated by the teacher model surpass the predictions of the student model, ensuring that the student model can learn to improve via reliable pseudo labels. Comparing the visualized results, it becomes evident that the proposed self-adaption pseudo label assigner can generate a greater quantity of higher-quality pseudo labels, but there is still have obvious gap in quality between the pseudo labels and the ground truth ones, especially in dense or small target scenarios, such as ships. Even so, this framework effectively mitigates the domain bias of the detector by establishing alignment at the label level.

3) Detection Results: We present the visualization results of the remote sensing teacher on all domain adaptation scenarios, accompanied by baseline detector [43], domain adaptive faster R-CNN for object detection in the wild (DAF) [6], and adversarial query transformers for domain adaptive object detection (AQT) [36]. As shown in Fig. 5, in rows 1–3 (xView → DOTA), our method showcases more accurate detection results for various categories. It effectively mitigates the occurrence of false positives generated by deformable-DETR and identifies challenging objects that the base detector might overlook. In rows 4 and 5 (UCAS-AOD → CARPK, CARPK → UCAS-AOD), our approach exhibits higher recall for detecting inconspicuous features and small objects. In row 6 (HRRSD → SSDD), this indicates that the remote sensing teacher successfully mitigates the domain shift problem in the UDA transformer detector when confronted with intricate cross-modalities scenarios. In contrast, other transformer-type methods only achieve limited improvements. This underscores the robust generalization capabilities of the remote sensing teacher. The visual results are consistent with the numerical assessment results, indicating the superior performance and generalization capabilities of the remote sensing teacher across multiple widely adopted domain adaptation scenarios in remote sensing.

V. Conclusion

This article introduces remote sensing teacher, an unsupervised cross-domain DETR designed for remote sensing scenarios. This is the first work that explores a DETR-based detector for tasks involving domain adaptation in the field of remote sensing. A novel LFA module is devised to help the detector effectively extract domain-invariant features in remote sensing scenes by achieving cross-domain feature alignment in the frequency space. Within the alignment module, the FMC mechanism ingeniously enhances the global feature representations of sparse objects in the low-level features with fewer computations. The LFFN selectively aligns the prominent features in noisy backgrounds by utilizing learnable filters with a gated mechanism. In addition, the remote sensing teacher employs a SPA that can automatically adjust the class-wise confidence threshold according to the model's learning status, thereby enabling the generation of high-quality pseudo-labels in scenarios with a long-tailed distribution. Leveraging these pseudo-labels further mitigates the domain bias of the detector by establishing alignment at the label level. Extensive experiments over multiple remote sensing domain adaptation scenarios show that remote sensing teacher achieves superior performance. This confirms the effectiveness and generalization ability of the proposed method. In future work, we plan to explore methods to improve the performance of cross-domain object detection using a limited quantity of samples or in the test-time stage.

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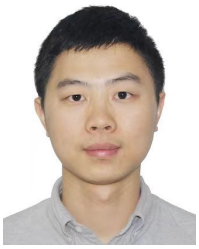
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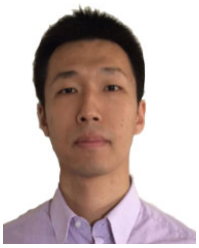
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